Turning Effects of Forces Main Note

Moment

Turning effect of force ⇒ moment
Lever ⇒ rigid bar resting on pivot or fulcrum
→ used to move a load when a force is applied

When force is applied ⇒ load experiences a turning force.
Rigid bar ⇒ is hinged at the pivot ⇒ can make a sweeping movement only.

Turning effect of a force is proportional to:

- Perpendicular distance form the pivot where the force is applied
- magnitude of force applied

Turning effect of a force ⇒ also known as *torque*

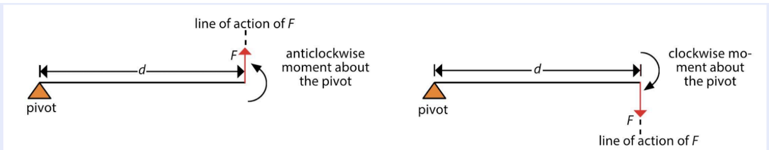
Definition: Moment of a force, M or torque, about a pivot is the product of the force F and the perpendicular distance d from the pivot to the line of action of the force
#PhysicsDefinitions

$$M = Fd$$

where F = force applied (N)
d = perpendicular distance from the pivot to the line of action of the force (m)

SI unit N m
⇒ it is a *vector* quantity. Moment of force has direction
a clockwise rotation about pivot ⇒ is a clockwise moment. Anticlockwise rotation about pivot ⇒ is anticlockwise moment

Important ⇒ in order to create moment ⇒ line of action of force must not pass through pivot



Principle of Moments

Center of Gravity

Stability
